

CARD8 inflammasome activation during HIV-1 cell-to-cell transmission

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eLife Assessment

Following up on their previous work, the authors investigated whether cell-to-cell transmission of HIV-1 activates the CARD8 inflammasome in macrophages. This is **important** given that inflammasome activation in myeloid cells triggers proinflammatory cytokine release. The data are **solid** and support the idea that CARD8 is activated by the viral protease and promotes inflammation. However, time-course analyses in primary T cells and macrophages and further information on the specific inflammasome involved would further increase the significance of the study.

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Abstract

Our previous work demonstrated that CARD8 detects HIV-1 infection by sensing the enzymatic activity of the HIV protease, resulting in CARD8-dependent inflammasome activation (Kulsuptrakul et al., 2023). CARD8 recognition of HIV-1 protease activity is conferred by a HIV protease substrate mimic within the CARD8 N-terminus, which when cleaved by HIV protease triggers CARD8 inflammasome activation. Here, we sought to understand CARD8 responses to HIV-1 when the virus is transmitted through cell-to-cell infection from infected cells to target cells via a viral synapse. We observed that cell-to-cell transmission of HIV-1 induces CARD8 inflammasome activation in immortalized cells and primary human monocyte-derived macrophages in a manner that is dependent on viral protease activity and largely independent of the NLRP3 inflammasome. Additionally, to further evaluate the viral determinants of CARD8 sensing, we tested a panel of HIV protease inhibitor resistant clones to establish how variation in HIV protease affects CARD8 activation. We identified mutant HIV-1 proteases that differentially cleave and activate CARD8 compared to wildtype HIV-1, thus indicating that natural variation in HIV protease affects not only the cleavage of the viral Gag-Pol polyprotein but also likely impacts innate sensing and inflammation.

Introduction

HIV-1 disease progression is characterized by chronic inflammation, immune activation, CD4⁺ T cell depletion and eventual destruction of the immune system and susceptibility to opportunistic infections. The primary cellular targets of HIV-1 are activated CD4⁺ T helper cells, specialized CD4⁺ T cell subtypes such as Th17 cells (Brenchley et al., 2008 [↗](#); Gosselin et al., 2009 [↗](#); Rodriguez-Garcia et al., 2014 [↗](#)), central memory cells (Chun et al., 1997b [↗](#), 1997a [↗](#), 1995 [↗](#)), and macrophages (Collman et al., 1990 [↗](#), 1989 [↗](#)). Chronic immune activation is primarily caused by rapid depletion of mucosal Th17 cells responsible for maintaining gut epithelial barrier integrity (Brenchley et al., 2008 [↗](#), 2006 [↗](#)). In addition to inflammation induced by circulating microbial ligands, inflammation can also originate from HIV-infected cells through activation of innate immune sensors that form cytosolic immune complexes known as inflammasomes.

Inflammasome activation ultimately results in activation of pro-inflammatory caspase 1 (CASP1), processing of inflammatory cytokines, and activation of a pore-forming protein belonging to the gasdermin superfamily called gasdermin D (GSDMD), which forms small pores in the cell membrane and initiates a lytic form of cell death known as pyroptosis, releasing mature inflammatory cytokines interleukin (IL)-1 β and IL-18 (Broz and Dixit, 2016 [↗](#); Fink and Cookson, 2005 [↗](#)).

In prior work, we and others showed that the inflammasome-forming sensor CARD8 senses HIV-1 infection through the detection of HIV-1 protease (HIV^{PR}) activity (Clark et al., 2022 [↗](#); Kulsuptrakul et al., 2023 [↗](#); Wang et al., 2021 [↗](#)). While the canonical function of HIV^{PR} is to cleave viral polyproteins during virion maturation, active HIV^{PR} is also released into the host cell, which is sensed by CARD8 via HIV^{PR} cleavage of its N-terminus and subsequent inflammasome activation. In this way, the CARD8 N-terminus functions as a “molecular tripwire” to recognize the enzymatic activity of HIV^{PR} and other viral proteases (Castro and Daugherty, 2023 [↗](#); Nadkarni et al., 2022 [↗](#); Tsu et al., 2023 [↗](#)). Moreover, HIV^{PR} cleavage of CARD8 occurs rapidly after infection such that HIV^{PR} inhibitors and fusion inhibitors, but not reverse transcriptase inhibitors can prevent CARD8 inflammasome activation, implying that CARD8 detects HIV-1 viral protease activity of virion-packaged HIV^{PR} or “incoming” HIV^{PR} upon virion fusion (Kulsuptrakul et al., 2023 [↗](#); Wang et al., 2024 [↗](#), 2021 [↗](#)). Interestingly, CARD8 inflammasome activation in resting CD4⁺ T cells results in pyroptosis but not the release of pro-inflammatory cytokines IL-1 β or IL-18 (Wang et al., 2024 [↗](#)), suggesting that the activation of CARD8 in T cells does not directly contribute to chronic inflammation. Here, we address whether or not CARD8 may influence HIV-1 pathogenesis through the maturation and release of IL-1 β from infected macrophages.

HIV-1 can be transmitted from one cell to another via two main mechanisms: “cell-free” infection through binding of free HIV-1 virions to target cells, and cell-to-cell infection whereby infected cells directly transfer virus to an uninfected target cell via the formation of a transient viral synapse (Chen et al., 2007 [↗](#); Galloway et al., 2015 [↗](#); Iwami et al., 2015 [↗](#)). Cell-to-cell transmission of HIV-1 has been reported between multiple HIV-1 target cell types including between active and resting CD4⁺ T cells (Agosto et al., 2018 [↗](#); Martin et al., 2010 [↗](#)) and between CD4⁺ T cells and macrophages (Baxter et al., 2014 [↗](#); Dupont and Sattentau, 2020 [↗](#); Lopez et al., 2019 [↗](#)). Cell-to-cell transmission delivers a large influx of virus to target cells, resulting in a high multiplicity of infection (MOI) (Agosto et al., 2015 [↗](#); Del Portillo et al., 2011 [↗](#); Duncan et al., 2013 [↗](#); Russell et al., 2013 [↗](#)), which has been proposed to enhance viral fitness by overwhelming host restriction factors including Tetherin/BST-2 (Jolly et al., 2010 [↗](#); Zhong et al., 2013 [↗](#)), SAMHD1 (Xie et al., 2019 [↗](#)), and TRIM5 α (Richardson et al., 2008 [↗](#)), evading adaptive immune responses including broadly neutralizing antibodies (Abela et al., 2012 [↗](#); Dufloo et al., 2018 [↗](#)). Cell-to-cell spread of HIV-1 is thus an important consideration in studying CARD8 inflammasome activation.

Here, we investigate both host and viral determinants of CARD8 inflammasome activation upon HIV-1 infection. We evaluate CARD8 sensing of HIV^{PR} during cell-to-cell transmission of HIV-1 to myeloid cells and find that CARD8 inflammasome activation occurs in the context of cell-to-cell transmission to both THP-1 cells, an acute myeloid leukemia cell line, and in primary monocyte-derived macrophages. Our findings suggest that CARD8 sensing of HIV^{PR} during cell-to-cell transmission of HIV-1 to macrophages may be a source of inflammatory cytokines that promote pathogenic chronic inflammation and disease progression. In addition, we also show that natural variation in HIV^{PR} due to resistance to protease inhibitors also affects CARD8 cleavage and subsequent inflammasome activation. Our results extend the role of incoming HIV^{PR} on CARD8-dependent activation of inflammasome responses as a function of cell type, mode of transmission, and virus evolution in response to antiviral therapy.

Results

Cell-to-cell transmission of HIV-1 induces CARD8 inflammasome activation

Our previous work investigating HIV-dependent CARD8 inflammasome activation used the cationic polymer DEAE-dextran, which is a common reagent to enhance viral infection in cell culture (Bailey et al., 1984 [DOI](#)). However, we occasionally found that DEAE-dextran could induce inflammasome activation in the absence of viral infection. To formally evaluate this, we first assessed whether or not we could observe CARD8-dependent inflammasome activation in a cell-free infection system in the absence of DEAE-dextran. We infected either wildtype (WT) or *CARD8* knockout (KO) THP-1 cells with HIV-1_{LAI} in either the presence or absence of DEAE-dextran and measured cell death and IL-1 β secretion 24 hours post-infection as readouts of inflammasome activation. We found that despite achieving similar levels of infection (20-30%) as measured by intracellular p24^{gag} after spinoculation with and without DEAE-dextran (**Figure 1A**, [left](#) [DOI](#)), we only detected robust CARD8-dependent inflammasome activation in WT THP-1 cells infected in the presence of DEAE-dextran (**Figure 1A** [DOI](#), middle and right). These data suggest that cationic polymer is necessary to observe HIV-dependent CARD8 inflammasome activation in our cell-free system.

These results prompted us to establish other models of HIV-1 infection and subsequent inflammasome activation that lack cationic polymers. Thus, we designed an *in vitro* coculture infection system to mimic HIV-1 cell-to-cell transmission by infecting SUPT1 cells, a T-cell lymphoma line (i.e., donor cells) and then mixing them with uninfected THP-1 cells (i.e., target cells). We opted for SUPT1 cells as the viral producer cell line because they are permissive to HIV-1 infection, and unlike THP-1 cells, SUPT1 cells do not respond to a known CARD8 inflammasome activator, Valbopro (VbP), as assayed by both IL-1 β secretion and cell death, indicating that SUPT1 cells do not have a functional CARD8 inflammasome pathway (**Figure 1B** [DOI](#), **Figure 1-figure supplement 1A** [DOI](#)). This allowed us to infer that inflammasome outputs (i.e. IL-1 β secretion) in our coculture system occur upon cell-to-cell transmission of HIV-1 from SUPT1 cells to the CARD8-competent THP-1 cells.

We found that coculture of THP-1 cells with HIV-1_{LAI}-infected but not mock-infected SUPT1 cells results in robust inflammasome activation as indicated by IL-1 β secretion, suggesting that our coculture system, which lacks DEAE-dextran, can induce HIV-dependent inflammasome activation via cell-to-cell infection (**Figure 1D** [DOI](#)). To further test this assumption, we prevented cell-to-cell contact using a virus-permeable transwell with a 0.4 μ m pore insert (**Figure 1C** [DOI](#)). In contrast to the cell-to-cell condition, HIV-1_{LAI}-infected SUPT1 cells (upper chamber) co-cultured with THP-1 cells (bottom chamber) did not lead to detectable IL-1 β secretion (**Figure 1D** [DOI](#)) despite equivalent amounts of free infectious virus as measured by reverse transcriptase level in the supernatant from the lower chamber (**Figure 1-figure supplement 1B** [DOI](#)). However, transmission of HIV-1

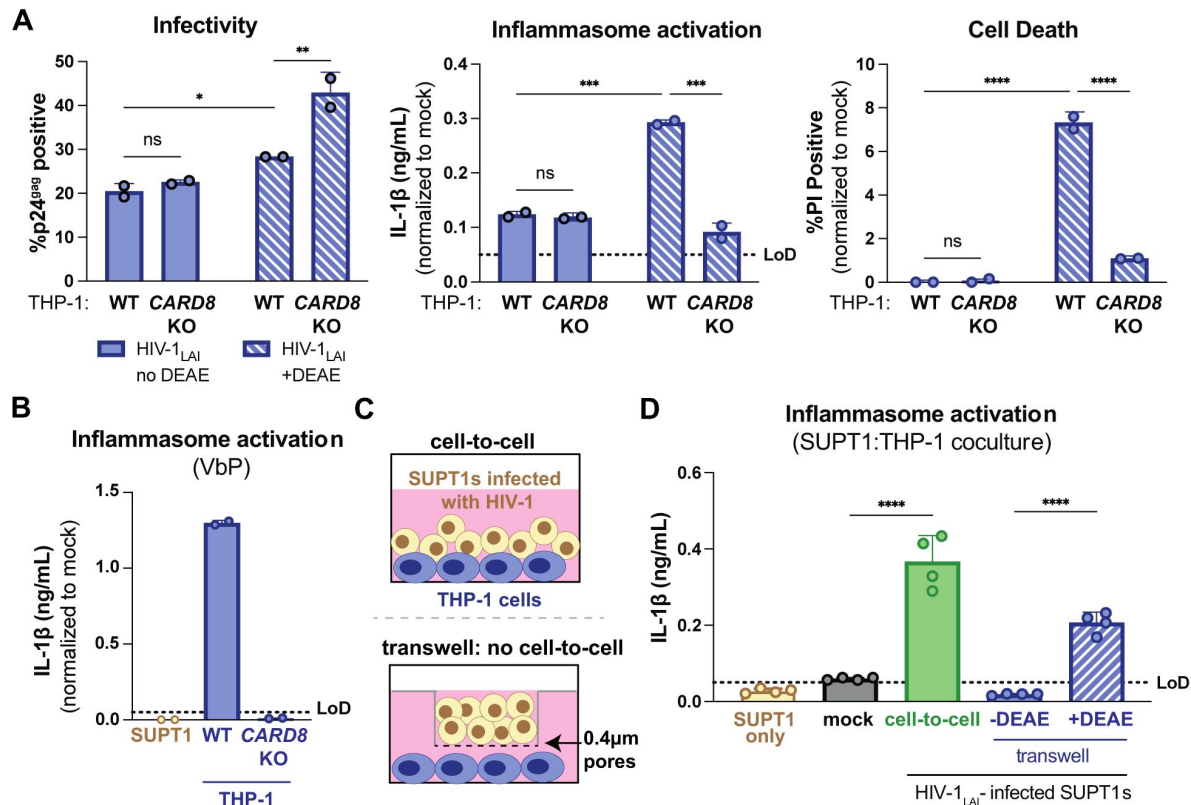


Figure 1.

HIV-1 cell-to-cell infection induces inflammasome activation.

(A) Wildtype (WT) or *CARD8* knockout (KO) THP-1 cells were infected with HIV-1_{LAI} at the same MOI in the presence or absence of DEAE-dextran (10 μ g/mL) then harvested after 24 hours and assayed for: left) percent infection via intracellular p24^{agg}, middle) inflammasome activation by IL-1 β secretion via IL-1R reporter assay, and right) cell death via propidium iodide (PI) dye uptake using flow cytometry. (B) SUPT1 or THP-1 cells were primed with Pam3CSK4 (500ng/mL) overnight then treated with 5 μ M Valbopro (VbP) for 24 hours then assessed for IL-1 β secretion as in (A). (C) Schematic illustrating experimental setup for SUPT1:THP-1 cell coculture. (D) THP-1 cells were primed overnight with Pam3CSK4 prior to coculture. SUPT1 cells were either mock infected or infected with HIV-1_{LAI} then cocultured with primed WT THP-1 cells 20 hours post infection. Mock- or HIV-1_{LAI}-infected SUPT1 cells were either mixed with the THP-1 cells or put in a transwell with a virus-permeable membrane in the presence or absence of DEAE-dextran. DEAE-dextran was added to the mock but not the mixed condition. Supernatant from the cocultures were harvested 3 days after starting the coculture and assayed for IL-1 β secretion as in (A) and (B). Dotted line indicates limit of detection (LoD). Datasets represent mean $\pm$ SD (A-B: n=2; D n=4 biological replicates). A-B: Two-way ANOVA with Tukey's test D: One-way ANOVA with Tukey's test using GraphPad Prism 10. ns = not significant, *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

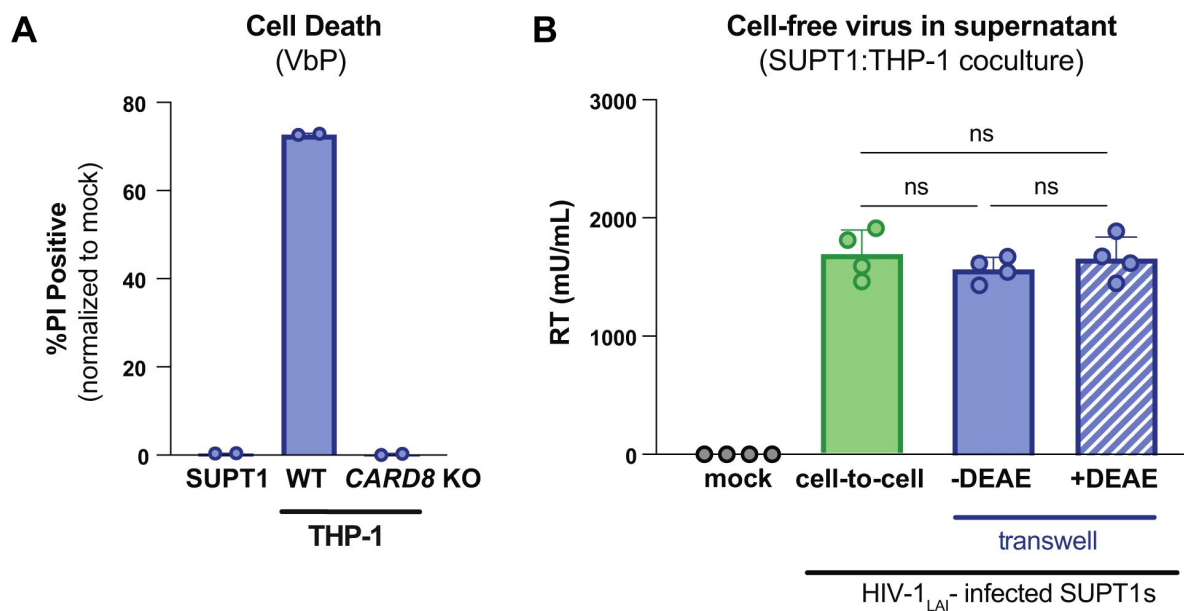


Figure 1- figure supplement 1.

HIV-1 cell-to-cell infection induces inflammasome activation.

(A) SUPT1 or THP-1 cells were primed with Pam3CSK4 (500ng/mL) overnight then treated with 5 μ M Valbopro (VbP) for 24 hours then assessed for cell death via propidium iodide (PI) uptake. (B) SUPT1 cells were cocultured with THP-1 cells as in Figure 1D and supernatant in the cell-to-cell condition and in the supernatant outside of the transwell was sampled and measured for cell-free infectious HIV virions via reverse transcriptase (RT) assay. Datasets represent mean $\pm$ SD (A: n=2; B: n=4 biological replicates). B: Two-way ANOVA with Tukey's test using GraphPad Prism 10. ns = not significant

through the transwell from infected SUPT1 cells to THP-1 cells and subsequent inflammasome activation was observed when media was supplemented with DEAE-dextran (**Figure 1D**), indicating that in the absence of cationic polymer, inflammasome activation in our SUPT1:THP-1 coculture system is triggered following HIV-1 cell-to-cell transmission.

We next assessed the role of CARD8 and other inflammasome sensors during cell-to-cell transmission of HIV-1. We cocultured mock- or HIV-1_{LAI}-infected SUPT1 cells with either WT or *CARD8* KO THP-1 cells and compared inflammasome activation by measuring levels of secreted IL-1 β . HIV-1_{LAI}-infected SUPT1 cells cocultured with WT but not *CARD8* KO THP-1 cells resulted in a significant increase in IL-1 β (**Figure 2A**). These results suggest that CARD8 is the primary sensor that drives inflammasome activation in HIV-1 cell-to-cell transmission to THP-1 cells. Since the NLRP3 inflammasome has previously been implicated in HIV-dependent inflammasome activation in CD4⁺ T cells (Zhang et al., 2021), we also assessed the effects of the NLRP3 inflammasome-specific inhibitor MCC950 (Coll et al., 2015; Primiano et al., 2016) on inflammasome activation in our coculture system. Treatment with MCC950 or the CASP1 inhibitor VX765 (Wannamaker et al., 2007) were sufficient to abrogate inflammasome activation induced by the ionophore nigericin, a well-characterized NLRP3 agonist (**Figure 2-figure supplement 1A**). However, in the HIV-1 coculture system, MCC950 treatment had only a modest effect on inflammasome activation while VX765 and the HIV^{PR} inhibitor lopinavir (LPV), which prevents CARD8 cleavage by HIV-1^{PR} (Kulsuptrakul et al., 2023; Wang et al., 2021), completely abrogated IL-1 β secretion (**Figure 2B**). We observed similar results during cell-free infection of THP-1 cells in the presence of DEAE-dextran (**Figure 2-figure supplement 1B**). Taken together, these findings indicate that HIV-dependent inflammasome activation is CARD8-dependent and largely NLRP3-independent.

CARD8 is required for inflammasome activation during HIV-1 cell-to-cell transmission into primary monocyte-derived macrophages

We next examined inflammasome activation upon HIV-1 cell-to-cell transmission in primary human monocyte-derived macrophages (MDMs). Previously, we had observed that CARD8 could sense active HIV-1^{PR} released into the host cytosol following viral fusion, which we refer to as “incoming” HIV-1^{PR} in our cell-free infection system in THP-1 cells using DEAE-dextran and spinoculation (Kulsuptrakul et al., 2023). Thus, we assessed the importance of viral entry by coculturing MDMs from three independent healthy donors with mock-, HIV-1_{LAI}-, or HIV-1_{NL4.3-BaL}-infected SUPT1 cells expressing CCR5 (SUPT1-CCR5) and assayed for inflammasome activation (**Figure 3A**). HIV-1_{LAI} is a CXCR4 tropic strain unable to infect macrophages whereas HIV-1_{NL4.3-BaL} is a CCR5 and macrophage-tropic strain. We observed inflammasome activation, as measured by IL-1 β secretion, in MDMs cocultured with HIV-1_{NL4.3-BaL}-infected SUPT1 cells but not in MDMs cocultured with mock- or HIV-1_{LAI}-infected SUPT1-CCR5 cells (**Figure 3A**). This demonstrates that HIV-dependent inflammasome activation can occur in MDMs during cell-to-cell infection in a manner dependent on viral entry. To further ascertain if this activation was CARD8-dependent and driven by incoming HIV^{PR} during SUPT1:MDM cell-to-cell transmission, we investigated the effects of different inhibitors on activation in MDM cocultures with HIV-1_{NL4.3-BaL}-infected SUPT1s. We observed that IL-1 β secretion was abrogated by treatment with lopinavir and VX765, indicating that inflammasome activation in MDM cocultures is dependent on HIV^{PR} and CASP1, respectively (**Figure 3A**). In addition, we used a reverse transcriptase inhibitor, nevirapine (NVP), to prevent synthesis of *de novo* translated HIV^{PR}, and thus any CARD8-dependent IL-1 β secretion would only be due to incoming HIV^{PR} in the presence of NVP. Indeed, we observed HIV-dependent inflammasome activation in the presence of NVP treatment that was added at the time of coculture, indicating that incoming HIV^{PR} is sufficient to elicit an inflammasome response (**Figure 3A**). Lastly, MDM cocultures treated with MCC950, an inhibitor of the NLRP3 inflammasome, had no effect on IL-1 β secretion (**Figure 3A**). Thus, cell-to-cell contact of infected cells with primary monocyte-derived macrophages can elicit an inflammasome response in a manner that is dependent on viral entry, CASP1, and incoming HIV^{PR}, and independent from NLRP3. Taken

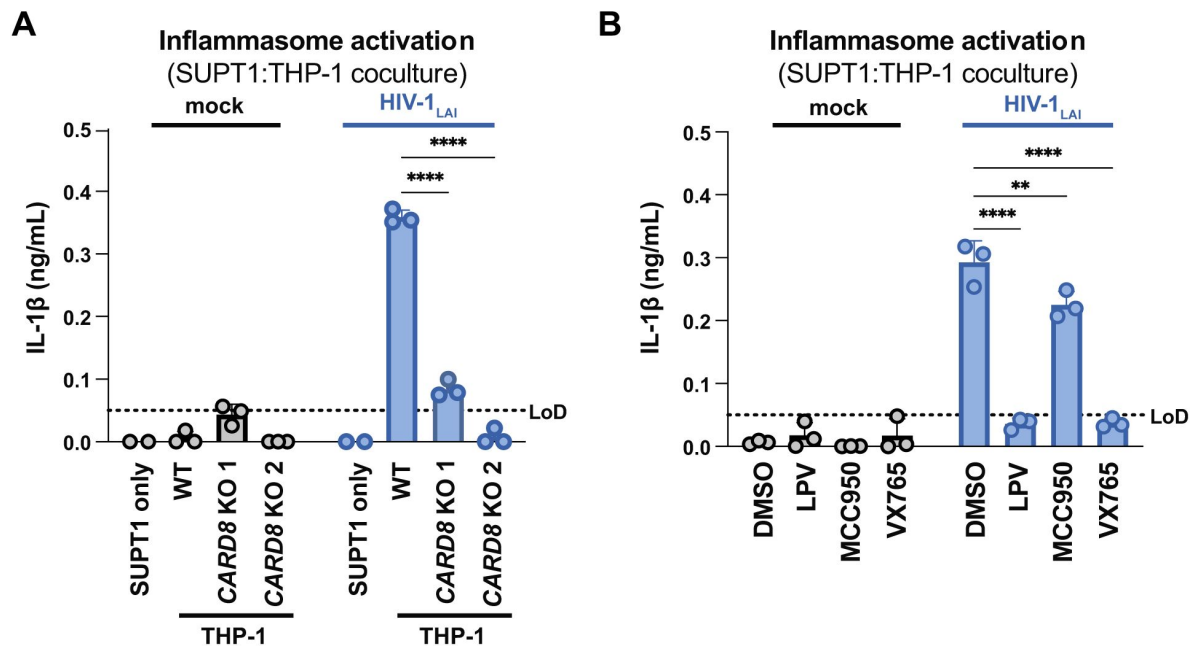


Figure 2.

HIV-1 cell-to-cell transmission induces CARD8-dependent activation largely independent of NLRP3.

(A) SUPT1 cells were either mock-infected or infected with HIV-1_{LAI} for 18-20 hours prior to coculture with wildtype (WT) or *CARD8* knockout (KO) THP-1 cells. The coculture was harvested 72 hours later to probe for IL-1 β secretion in the coculture supernatant via IL-1R reporter assay. THP-1 cells were primed with Pam3CSK4 (500ng/mL) for 16-24 hours prior to coculture. SUPT1 cells were infected with HIV-1_{LAI} such that 30% of the cells were positive for intracellular p24^{gag} after 18-20 hours. (B) SUPT1 cells were either mock or HIV-1_{LAI}-infected as in (A) for 18-20 hours then incubated in DMSO, lopinavir (LPV), MCC950, or VX765 at 0.01%, 5 μ M, 10 μ M, or 1 μ g/mL, respectively, for 15 minutes prior to coculturing with primed WT THP-1 cells. The coculture was assessed for subsequent inflammasome activation after 72 hours as in (A). Dotted line indicates limit of detection (LoD). Datasets represent mean $\pm$ SD (n=3 biological replicates). Two-way ANOVA with Dunnett's test using GraphPad Prism 10. ns = not significant, *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

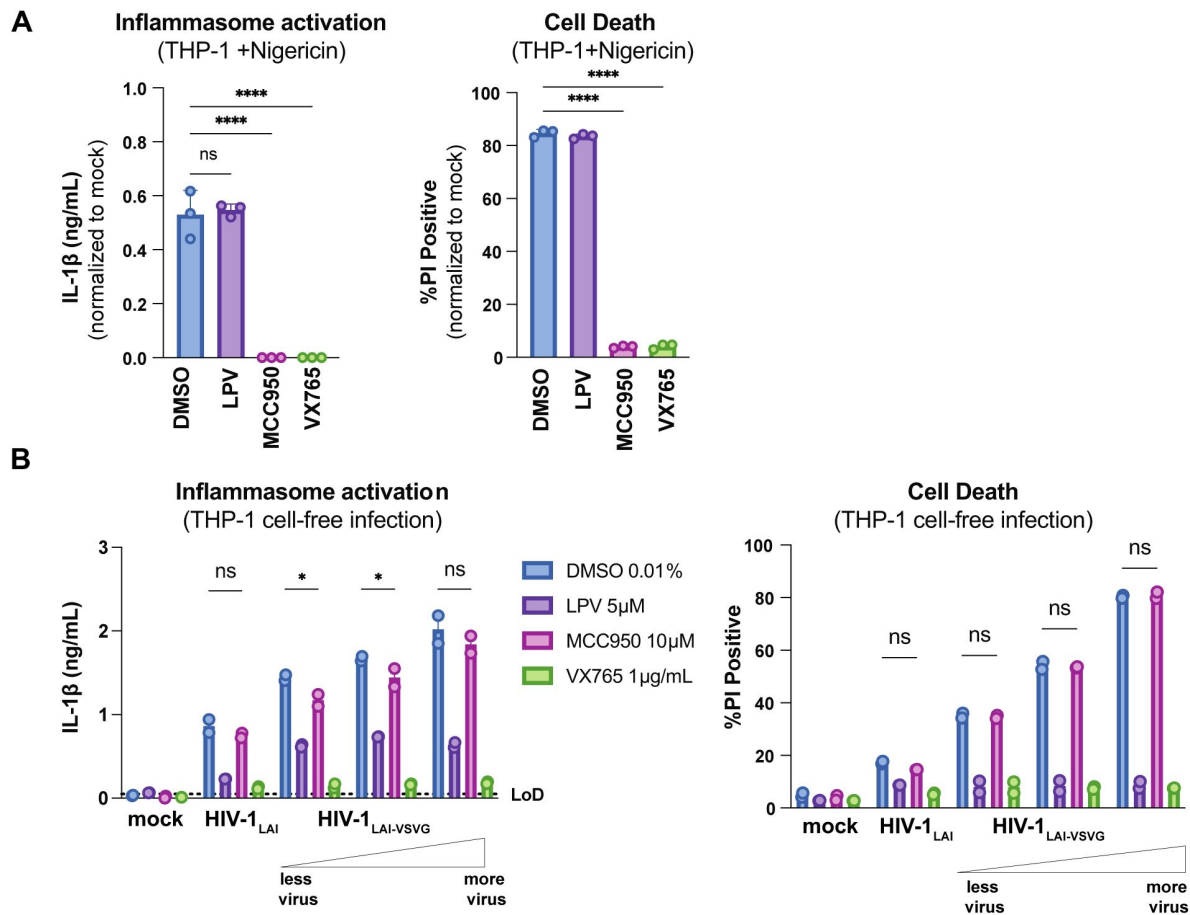


Figure 2-figure supplement 1.

HIV-dependent inflammasome activation is largely NLRP3- independent.

(A) Wildtype THP-1 cells were pre-treated with either DMSO, lopinavir (LPV), MCC950, or VX765 at 0.01%, 5 μ M, 10 μ M, or 1 μ g/mL, respectively, for 15 minutes prior to 4-hour treatment with 5 μ g/mL nigericin. Subsequent inflammasome activation was assessed via (A-left) IL-1 β secretion via IL-1R reporter assay and (A-right) cell death via propidium iodide (PI) dye uptake. (B) Wildtype THP-1 cells were pre-treated with indicated inhibitors as in (A) then infected with either HIV-1_{LAI} or increasing amounts of VSV-G pseudotyped HIV-1_{LAI} (HIV-1_{LAI-VSVG}) in the presence of 10 μ g/mL DEAE-dextran. Subsequent inflammasome activation was assessed 24 hours post infection via IL-1 β secretion and cell death as in (A). Dotted line indicates limit of detection (LoD). Datasets represent mean $\pm$ SD (A: n=3, B: n=2 biological replicates). One-way (A) or Two-way (B) ANOVA with Dunnett's test using GraphPad Prism 10. ns = not significant, *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

together, these data suggest that in the context of cell-to-cell transmission, CARD8 is likely the inflammasome-forming sensor that detects HIV-1 infection via incoming HIV^{PR} activity in primary macrophages.

To more definitively probe the role of CARD8 in HIV-1 induced inflammasome activation in MDMs, we genetically edited MDMs by isolating monocytes from three healthy donors and electroporating them with Cas9 RNPs complexed with three unique sgRNAs per gene targeting either *AAVS1*, a safe harbor locus, or *CARD8*. Edited MDMs were then differentiated for 6 days prior to evaluating KO efficiency and initiating cocultures with HIV-1-infected SUPT1 cells. In all three donors, we observed a marked reduction of the full-length and FIIND-processed CARD8 protein by immunoblotting with an antibody that detects the CARD8 C-terminus in *CARD8* KO MDMs relative to the *AAVS1* KO control (**Figure 3B**). In addition, we confirmed both *CARD8* and *AAVS1* KO at the genetic level via Synthego ICE analysis (Conant et al., 2022), measuring >85% KO efficiency (**Figure 3B**). We also observed robust inflammasome activation upon treatment with CARD8 inflammasome activator VbP as measured by IL-1 β secretion in *AAVS1* KO MDMs from 2 of the 3 donors, which was completely abrogated in *CARD8* KO MDMs, confirming functional loss of CARD8 (**Figure 3C**). We next cocultured either *AAVS1* KO or *CARD8* KO MDMs with mock or HIV-1_{NL4.3-BaL}-infected SUPT1-CCR5 cells at a 1:1 ratio and measured inflammasome activation via IL-1 β secretion 48 hours post-coculture. In all 3 donors, we observed significant reduction in inflammasome activation in *CARD8* KO cocultures relative to the *AAVS1* KO control (**Figure 3D**). Taken together, these data demonstrate that CARD8 is required for inflammasome activation in MDMs during HIV-1 cell-to cell transmission.

Protease inhibitor resistant strains of HIV-1 differentially cleave and activate CARD8

The consequences of CARD8 inflammasome activation on viral replication have been challenging to assess given that viral fitness is intrinsically linked to viral protease processing of the viral polyprotein such that inhibiting HIV^{PR} also prevents viral replication. In an attempt to circumvent this issue, we surveyed a panel of multi-HIV^{PR} inhibitor-resistant (PI-R) infectious molecular clones of HIV-1 (Varghese et al., 2013). This panel of PI-R molecular clones vary in resistance to HIV protease inhibitors including nelfinavir (NFV), fosamprenavir (FPV), saquinavir (SQV), indinavir (IDV), atazanavir (ATV), lopinavir (LPV), tipranavir (TPV), and darunavir (DRV). Each molecular clone encodes 4 to 11 mutations in HIV^{PR} as well as various compensatory HIV^{gag} mutations (Varghese et al., 2013) (**Table S1**).

We initially tested if PI-R HIV-1 proviruses differentially cleave CARD8 by co-transfecting HEK293T cells with an expression plasmid encoding an N-terminal mCherry tagged human CARD8 and either wildtype HIV-1_{LAI} or PI-R HIV-1 proviruses. HIV-1_{LAI} protease cleaves CARD8 between phenylalanine (F) 59 and F60 (Wang et al 2021), resulting in a ~33kDa product (**Figure 4A-top**). By quantifying the 33kDa CARD8 cleavage product with each HIV-1 provirus, we identified a PI-R clone that exhibited similar efficiency at cleaving CARD8 to HIV-1_{LAI} (i.e., PI-R1), PI-R clones that were markedly less efficient at cleaving CARD8 than HIV-1_{LAI} (i.e., PI-R2, PI-R3, PI-R5, PI-R9, and PI-R10) and two PI-R clones, PI-R12 and PI-R13, that were more efficient at cleaving CARD8 than HIV-1_{LAI} (**Figure 4A-top**, **Table S1**). Of note, all PI-R proviruses had similar levels of HIV^{PR} activity for HIV^{gag} polyprotein processing from p55^{gag} to p24^{gag} as indicated by the ratio of p24^{gag}/p55^{gag} quantified from the anti-p24^{gag} immunoblot (**Figure 4A-middle**). These results indicate that naturally occurring HIV-1 protease mutations can influence host targets like CARD8.

We next assessed if PI-R clones exhibiting reduced (PI-R2 and -9) or increased (PI-R12 and -13) cleavage of CARD8 relative to HIV-1_{LAI} (**Figure 4A**, **4B**, and **Table S1**) resulted in differential inflammasome activation. HEK293T cells endogenously express CARD8 but lack the downstream components (i.e., CASP1, GSDMD, and IL-1 β /IL18) of the inflammasome pathway. Thus, we reconstituted the inflammasome pathway in HEK293T cells by co-transfection of human

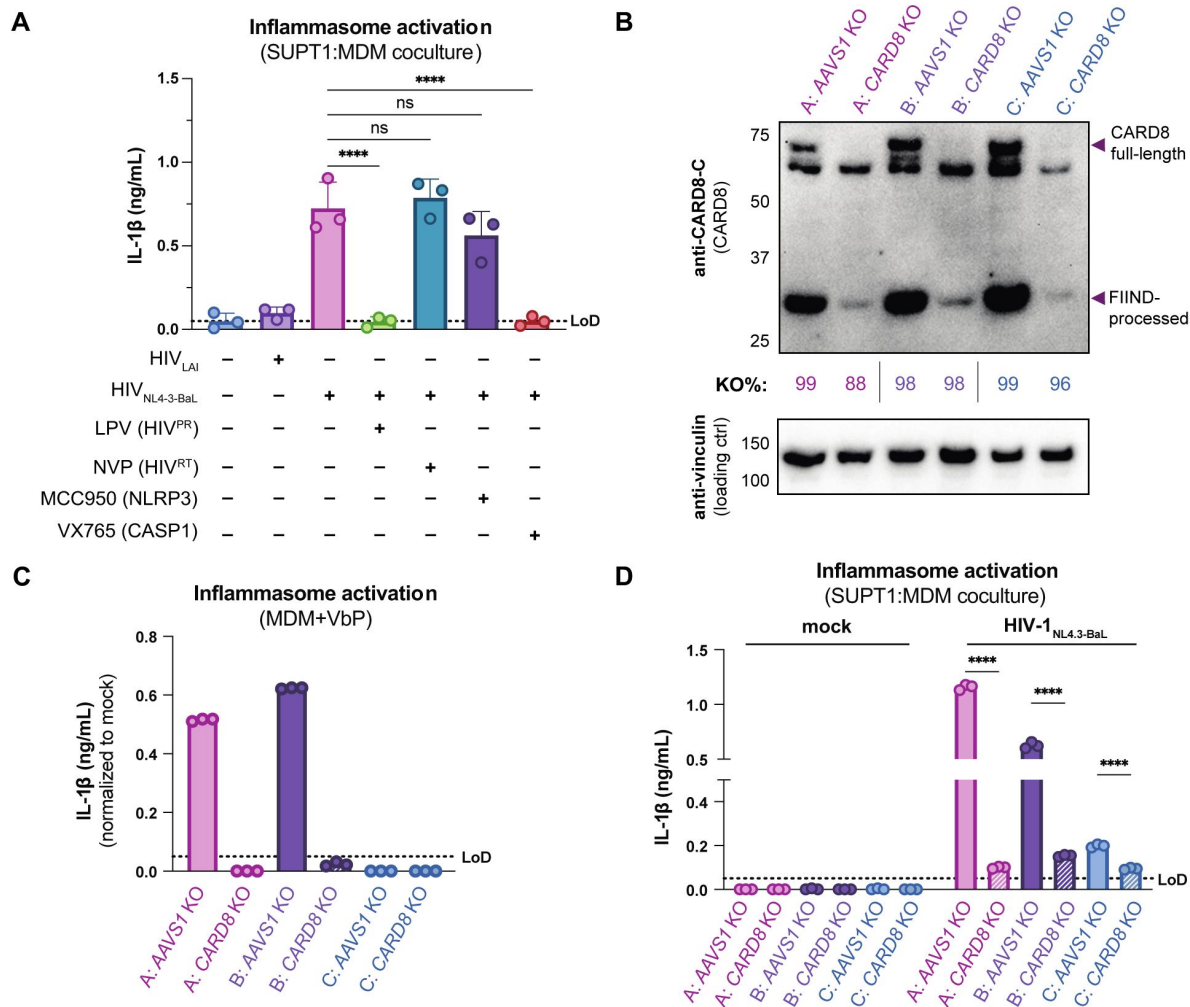


Figure 3.

Cell-to-cell HIV infection induces CARD8-dependent activation in monocyte-derived macrophages (MDMs).

(A) MDMs were cocultured with SUPT1 cells expressing CCR5 (SUPT1-CCR5) that were mock-, HIV-1_{LAI}- or HIV-1_{NL4.3-BaL}-infected then assayed for inflammasome activation 48 hours post-coculture for IL-1β secretion. Fifteen minutes before starting the coculture, SUPT1-CCR5 cells infected with HIV-1_{NL4.3-BaL} were pre-treated with either DMSO, lopinavir (5μM), nevirapine (50μM), MCC950 (10μM) or VX765 (1μg/mL), inhibiting HIV-1 protease (HIV^{PR}), HIV-1 reverse transcriptase (HIV^{RT}), NLRP3, or caspase 1 (CASP1), respectively. SUPT1-CCR5 cells were infected with HIV-1_{LAI} and HIV-1_{NL4.3-BaL} such that 2% and 7% of cells, respectively, were positive for intracellular p24^{gag} after 24 hours. (B) MDMs from 3 independent healthy donors were knocked out (KO) for either AAVS1 or CARD8 using a Synthego gene KO kit and were immunoblotted using an anti-CARD8 antibody or anti-vinculin. Full-length and FIIND-processed CARD8 intermediates are marked with a purple arrow. Table between CARD8 and vinculin blot shows Synthego gene KO% scores for each donor KO line. (C) AAVS1 and CARD8 KO MDM lines were primed with Pam3CSK4 (500ng/mL) overnight and then treated with ValboraPro (VbP) for 24 hours before assessing for IL-1β secretion. (D) AAVS1 and CARD8 KO MDM lines were cocultured with SUPT1-CCR5 cells mock-, or HIV-1_{NL4.3-BaL}-infected then assayed for inflammasome activation 48 hours post-coculture for IL-1β secretion. Dotted line indicates limit of detection (LoD). Datasets represent mean ± SD (A: n=3 independent donors, C-D: n=3 technical replicates per donor). (A) One-way or (D) two-way ANOVA with Tukey's test (using GraphPad Prism 10. ns = not significant, *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

Clone name	PI-resistance mutations in HIV ^{PR}	HIV ^{gag} mutation	Strongest PI-R ¹	% CARD8 cleavage (relative to HIV _{LAI})
HIV-1 _{LAI}	wildtype	wildtype	wildtype	100
CA126802 (PI-R1)	11I, 32I, 33F, 46I, 47V, 54M, 58E, 73S, 84V, 89V, 90M	A431V (NC/p1) I437N (NC/p1) P453LF (p1/p6)	FPV, LPV, TPV, DRV	51
CA122805 (PI-R2)	10F, 33F, 43T, 46L, 54V, 82A, 84V, 90M	A431V (NC/p1) R532S (p1/p6)	SQV	7
CA126805 (PI-R3)	33F, 43T, 46I, 48V, 50V, 54S, 82A	I437N (NC/p1) L449VF (p1/p6) PTAP insertion (p6)	SQV, IDV, LPV	8
CA96457 (PI-R5)	48V, 53L, 54V, 82A, 90M	K436R (NC/p1)	SQV	11
CA96458 (PI-R9)	10F, 30N, 33F, 43T, 84V, 88D, 90M	A431V (NC/p1)	NFV, SQV	24
CA50834-1 (PI-R10)	24I, 46I, 54V, 76V, 82T	A431V (NC/p1)	IDV, LPV	38
CA96451 (PI-R12)	32I, 33F, 43T, 46I, 47V, 54M, 73S, 82A, 89V, 90M	I437N (NC/p1) PTAP insertion (p6)	FPV, LPV	192
CA20392-1 (PI-R13)	24I, 46L, 54V, 82A	A431V (NC/p1)	LPV	206

Table S1

Protease inhibitor resistance mutations and relative CARD8 cleavage.

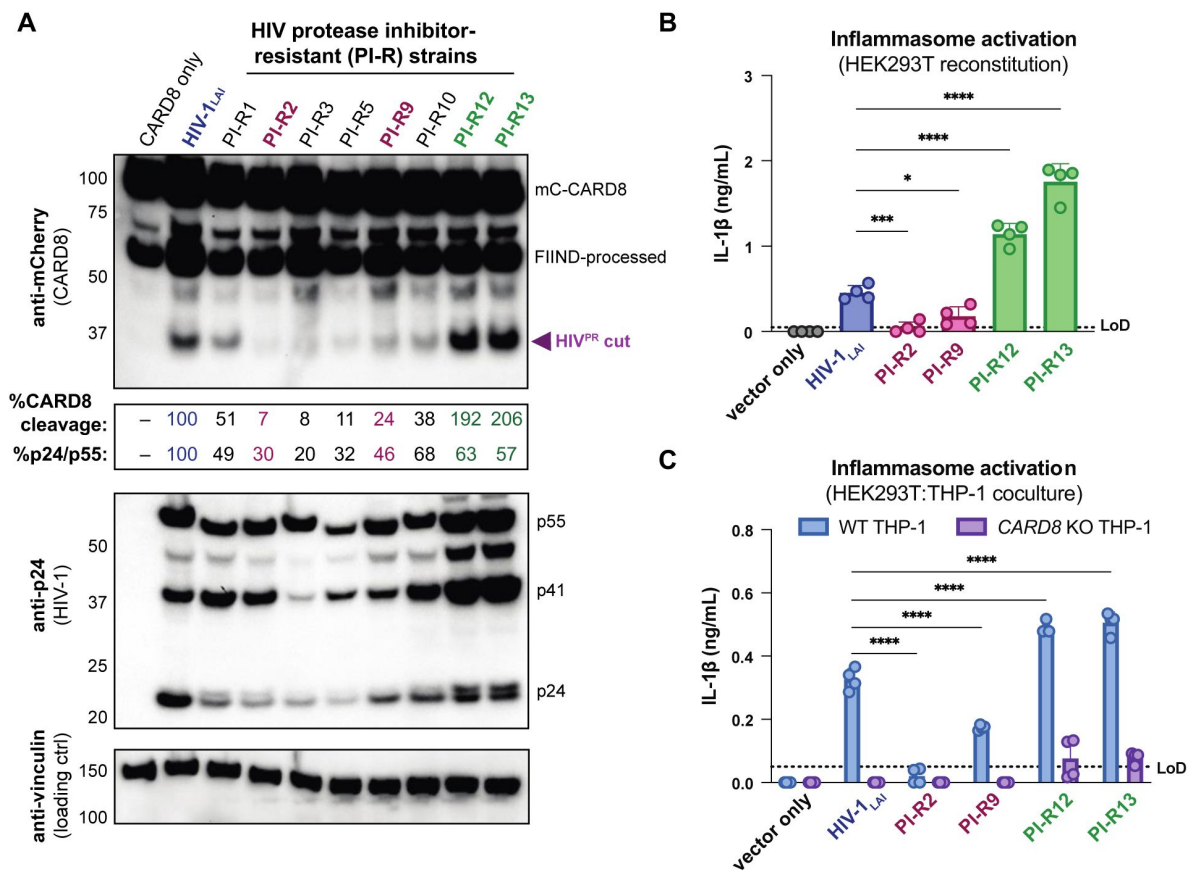


Figure 4.

Protease inhibitor resistant strains of HIV-1 differentially cleave and activate CARD8.

(A) HEK293T cells were transfected with a construct encoding CARD8 with an N-terminal mCherry tag (mCherry-CARD8) and indicated HIV-1 proviral constructs. Protease inhibitor-resistant (PI-R) clones of HIV-1 are a subset of a panel expressing prototypical multidrug resistant HIV-1 protease (HIV^{PR}) in an NL4.3 backbone (Table S1). Top: Immunoblotting using anti-mCherry antibody to detect mCherry-CARD8. The full-length (mc-CARD8) and FIIND-processed bands are indicated as well as the HIV^{PR} cut product. The band at ~45 kDa is the result of cleavage by the 20S proteasome. % CARD8 cleavage was calculated by quantifying the HIV^{PR} cut band relative to the HIV-1_{LA1} control using BioRad Image Lab 6. Middle: Immunoblotting with an anti-p24^{gag} antibody showing HIV-1^{gag} cleavage products p41^{gag} and p24^{gag}, and/or full-length HIV-1^{gag}, p55^{gag}. %p24/p55 was calculated from the ratio of p24^{gag} versus p55^{gag} product by quantifying the volume of the p24^{gag} bands versus the p55^{gag} band relative to the HIV-1_{LA1} control using BioRad Image lab 6. Bottom: Immunoblotting with an anti-vinculin antibody to detect vinculin as a loading control. (B) HEK293T cells were transfected with human caspase 1 and human pro-IL-1 β , and either carrier vector or indicated HIV-1 proviruses then probed for IL-1 β secretion 24 hours post-transfection via IL-1R reporter assay. (C) HEK293T cells were transfected with indicated HIV-1 proviruses (300ng). 24 hours post-transfection either wildtype (WT) or *CARD8* knockout (KO) THP-1s were overlaid on the transfected HEK293T cells in a 1:1 ratio. THP-1s were primed with Pam3CSK4 overnight prior to coculture. Supernatants were harvested 24 hours post coculture to assay for IL-1 β secretion as in (B). Dotted line indicates limit of detection (LoD). Datasets represent mean $\pm$ SD (n=4 biological replicates). p-Values were determined by two-way ANOVA with Dunnett's test using GraphPad Prism 10. ns = not significant, *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

caspase 1, human pro-IL-1 β , and either empty vector, HIV-1_{LAI} or representative PI-R proviruses then quantified CASP1-dependent processing of pro-IL-1 β as a readout of CARD8 inflammasome activation as in (Tsu et al., 2023 [\[1\]](#)). Consistent with the observed differences in CARD8 cleavage by PI-R clones (**Figure 4A** [\[1\]](#)), we found that PI-R2 and PI-R9, which exhibited less CARD8 cleavage than HIV-1_{LAI}, also induced lower IL-1 β levels than HIV-1_{LAI} (**Figure 4B** [\[1\]](#)). Similarly, PI-R12 and PI-R13, which demonstrated enhanced CARD8 cleavage, elicited higher IL-1 β levels than HIV-1_{LAI} (**Figure 4B** [\[1\]](#)). We next assessed inflammasome activation by the PI-R clones in a cell-to-cell transmission model using HEK293T cells as donor cells rather than SUPT1 cells and either WT or CARD8 KO THP-1 cells as the target line at a 1:1 ratio. We opted to overexpress the HIV-1_{LAI} or the PI-R proviruses in HEK293T cells rather than infecting SUPT1 cells due to dramatic variability in replication kinetics between PI-R strains. In these HEK293T:THP-1 cocultures, we observed that cell-to-cell transmission of PI-R2 and PI-R9 resulted in lower IL-1 β levels while PI-R12 and PI-R13 resulted in higher IL-1 β levels compared to HIV-1_{LAI}, respectively (**Figure 4C** [\[1\]](#)), consistent with our findings from CARD8 cleavage (**Figure 4A** [\[1\]](#)) and reconstituted inflammasome assays (**Figure 4B** [\[1\]](#)). Our findings suggest that HIV-dependent inflammasome activation is under genetic control of the viral protease in a manner that can be increased or decreased with naturally occurring mutations induced by drug resistance.

Table S1 [\[1\]](#) shows the protease inhibitor-resistant (PI-R) clones assayed in **Figure 4** [\[1\]](#) with corresponding mutations in HIV protease (HIV^{PR}) and HIV^{gag}. ¹These clones were previously cloned and assayed for PI-R in (Varghese et al., 2013 [\[1\]](#)). The PI-R subset used in **Figure 4B** [\[1\]](#) are bolded and highlighted in red or green and denote either hypo- or hyper-active CARD8 cleavage, respectively. NFV-nelfinavir; FPV-fosamprenavir; SQV-saquinavir; IDV- indinavir; LPV- lopinavir; TPV- tipranavir; DRV- darunavir. The consensus subtype B sequence can be found on the Stanford HIV Drug Resistance Database (HIVDB) ("Stanford - HIV Drug Resistance Database," n.d. [\[1\]](#)). Relative CARD8 cleavage was determined by quantifying band volume of the CARD8 cleavage product in BioRad Image Lab 6 and comparing to cleavage with HIV-1_{LAI}.

Discussion

We demonstrate that cell-to-cell transmission of HIV-1 to myeloid cells yields CARD8-dependent inflammasome activation via incoming HIV^{PR}. This activation occurs in a largely NLRP3-independent manner in these myeloid cell types despite being previously implicated as an innate HIV sensor in CD4⁺ T cells (Zhang et al., 2021 [\[1\]](#)). In addition, we identified protease inhibitor resistant strains of HIV-1 that differentially cleave and activate the CARD8 inflammasome. Thus, HIV^{PR} mutants selected for their resistance to different protease inhibitors also affect their ability to cleave host proteins including the inflammasome-forming sensor CARD8.

CARD8 as the primary innate sensor of HIV-1 infection

Previously, both the NLRP3 and IFI16 inflammasomes have been implicated as innate sensors of HIV-1 infection and drivers of CD4⁺ T cell depletion using blood and lymphoid- derived CD4⁺ T cells, respectively, and cell-to-cell transmission was reported to be crucial for IFI16 sensing of abortive HIV transcripts (Doitsh et al., 2014 [\[1\]](#); Galloway et al., 2015 [\[1\]](#); Monroe et al., 2014 [\[1\]](#); Zhang et al., 2021 [\[1\]](#)). However, the mechanism of activation of NLRP3 activation in response to HIV-1 remains elusive. Similarly, there have been reports that IFI16 is not an inflammasome-forming sensor, and instead a nuclear transcriptional regulator of antiviral genes including type I interferons and RIG-I (Hornung et al., 2009 [\[1\]](#); Jiang et al., 2021 [\[1\]](#); Thompson et al., 2014 [\[1\]](#)), suggesting that there may be other mechanisms of CD4⁺ T cell depletion and HIV- dependent inflammasome activation at play. Indeed, CARD8, which is expressed and functional in naïve and memory CD4⁺ and CD8⁺ T cells (Linder et al., 2020 [\[1\]](#)), was recently shown to be required for pyroptosis in primary human blood- and lymphoid-derived CD4⁺ T cells and humanized mouse models (Wang et al., 2024 [\[1\]](#)), implicating CARD8 as a major driver of CD4⁺ T cell depletion during

HIV-1 infection. In this study and our prior work (Kulsuptrakul et al., 2023 [DOI](#)), we demonstrate that CARD8 is also the primary innate sensor during HIV-1 infection in myeloid cell types. However, our present study does not rule out the possibility that under certain conditions or in certain cell types, NLRP3 activation may occur, for example following GSDMD pore formation following CARD8 inflammasome activation and play a more profound role in promoting HIV-dependent inflammation. Nonetheless, these data along with other recent work (Wang et al., 2024 [DOI](#)) strongly suggest that CARD8 is a major innate sensor of HIV-1 infection.

Protease inhibitor resistance mutations and inflammatory disease

Given the important role of HIV^{PR} in replication, early combination antiretroviral therapy for people living with HIV (PLWH) included protease inhibitors along with reverse transcriptase inhibitors. However, resistance mutations to protease inhibitors quickly arose in PLWH through mutations around the HIV^{PR} active site allowing for polyprotein processing and viral maturation while avoiding drug inhibition. Despite typically having poor overall viral fitness due to less efficient polyprotein processing and replication relative to wildtype HIV-1 in the absence of protease inhibitors, these mutant drug-resistant HIV-1 strains can persist in PLWH on antiviral therapy, posing a major threat to controlling disease progression (De Luca, 2006 [DOI](#); Martínez-Picado et al., 1999 [DOI](#); Prado et al., 2002 [DOI](#); Resch et al., 2002 [DOI](#)). To compensate for mutations in HIV^{PR} that change its substrate specificity, HIV^{gag} sometimes evolves mutations around HIV^{PR} cleavage sites to permit proper polyprotein processing (Varghese et al., 2013 [DOI](#)). Here, we identified multiple HIV^{PR} inhibitor-resistant strains of HIV-1 that can differentially cleave and activate the CARD8 inflammasome (Figure 4 [DOI](#), Table S1 [DOI](#)). As the degree of inflammation is a better predictor of disease progression in untreated individuals than viral load (Deeks et al., 2004 [DOI](#); Giorgi et al., 1999 [DOI](#)), we speculate that differential CARD8 inflammasome activation could influence disease progression for PLWH harboring HIV^{PR} resistance mutations that cleave CARD8 more or less efficiently. More broadly, we suggest that host targets of viral proteases like CARD8 may influence the selection of viral variants during treatment with antiviral protease inhibitor monotherapies.

Viral influx activates the CARD8 inflammasome

In this study, we demonstrate that HIV-dependent CARD8 inflammasome activation during cell-free infection requires a cationic polymer like DEAE-dextran (Figure 1A [DOI](#)). Previously, we have demonstrated that CARD8 senses incoming HIV^{PR} within 2 hours post-infection, killing infected cells well before p24^{gag} can be expressed. Despite being infected with the same amount of virus and exhibiting similar percent infection 24 hours post-infection, as measured by intracellular p24^{gag}, with and without DEAE-dextran, we speculate that DEAE-dextran during cell-free infection may increase the effective MOI or facilitate superinfection, leading to more efficient viral influx to trigger CARD8 sensing, and thus p24^{gag} positive cells after 24 hours may be an underestimate of total infected cells in the DEAE-dextran condition. On the other hand, cell-to-cell transmission of HIV-1 between infected donor cells cocultured with target cells at a 1:1 ratio is sufficient to induce CARD8-dependent activation in the absence of cationic polymer. We infer that this is also likely a product of the efficiency of viral entry and the necessity for multiple virions infecting at the same time to deliver a sufficient amount of active HIV^{PR} for cytosolic CARD8 sensing. We postulate that under certain physiological conditions, cell-to-cell transmission can cause CARD8 inflammasome activation when there is an influx of active incoming HIV^{PR} across the viral synapse. Taken together, we speculate that both cell-free infection facilitated by cationic polymer and cell-to-cell transmission can achieve sufficient levels of active HIV^{PR} influx to activate the CARD8 inflammasome.

Macrophages have been reported to be primarily infected through phagocytosis of infected CD4⁺ T cells or cell-to-cell transmission (Dupont and Sattentau, 2020 [DOI](#); Martínez-Méndez et al., 2017 [DOI](#); Orenstein, 2000 [DOI](#)). We demonstrate that unlike CD4⁺ T cells, which are rapidly depleted by HIV-1 infection and do not release IL-1 β or IL-18 (Linder et al., 2020 [DOI](#)), primary macrophages release

pro-inflammatory cytokines in response to HIV^{PR} during cell-to-cell infection (**Figure 3A** [↗](#), **Figure 3D** [↗](#)), thus representing a potential source of sustained IL-1 β and subsequent chronic immune activation. In addition to promoting chronic immune activation, HIV-dependent IL-1 β release from macrophages may also contribute to HIV-1 pathogenesis by activating nearby CD4⁺ T cells, rendering them susceptible to becoming infected with HIV-1, and thus indirectly promoting CD4⁺ T cell depletion. Collectively with our prior work ([Kulsuptrakul et al., 2023](#) [↗](#)), our findings provide further evidence that CARD8 inflammasome activation is driven by incoming HIV^{PR} under conditions where multiple virions may enter cells, and thus could be a potential driver of HIV-1 pathogenesis by promoting chronic immune activation.

Methods

Plasmids and Reagents

pMD2.G used for HIV-1_{LAI}-VSVG production was a gift from Didier Trono ([Addgene](#)). HIV-1_{LAI} has been previously described ([Peden et al., 1991](#) [↗](#)). The following reagents were obtained through the NIH HIV Reagent Program, Division of AIDS, NIAID, NIH: lopinavir (LPV), nevirapine (NVP), Human Immunodeficiency Virus 1 (HIV-1) NL4-3 BaL Infectious Molecular Clone (p20-36) (HIV-1_{NL4.3-BaL}), ARP-11442, contributed by Dr. Bruce Chesebro ([Chesebro et al., 1992](#) [↗](#), [1991](#) [↗](#); [Toohey et al., 1995](#) [↗](#); [Walter et al., 2005](#) [↗](#)), and Panel of Multi-Protease Inhibitor Resistant Infectious Molecular Clones, HRP-12740, contributed by Dr. Robert Shafer ([Varghese et al., 2013](#) [↗](#)). Mutant HIV^{PR} sequences were amplified from clinically-derived viral cDNA encoding protease genes with resistance to multiple PRIs then cloned into an NL4.3 backbone with overhangs including the 3' end of gag with the gag cleavage site and the 5' end of RT as previously described ([Varghese et al., 2013](#) [↗](#)). CARD8 variant constructs were cloned as previously described ([Kulsuptrakul et al., 2023](#) [↗](#)). VX765 and MCC950 were sourced from Invivogen (cat: inh-vx765i-1 and inh-mcc, respectively).

Cell culture

SUPT1 (ATCC) and THP-1 cells (ATCC) were cultured in RPMI (Invitrogen) with 10% FBS, 1% penicillin/streptomycin antibiotics, 10 mM HEPES, 0.11 g/L sodium pyruvate, 4.5 g/L D-glucose and 1% Glutamax. Primary monocytes were cultured in RPMI (Invitrogen) with 10% FBS, and 1% penicillin/streptomycin antibiotics and differentiated in the presence of 20ng/mL GM-CSF (Peprotech cat: 300-03) and 10ng/mL M-CSF (Peprotech cat: 300-25). HEK293T (ATCC) lines were cultured in DMEM (Invitrogen) with 10% FBS and 1% penicillin/streptomycin antibiotics. All lines routinely tested negative for mycoplasma bacteria (Fred Hutch Specimen Processing & Research Cell Bank).

HIV-1_{LAI}, HIV-1_{LAI}-VSVG, and HIV-1_{NL4.3-BaL} production

293T cells were seeded at $2\text{--}3 \times 10^5$ cells/well in six-well plates the day before transfection using TransIT-LT1 reagent (Mirus Bio LLC) at 3 μ L transfection reagent/well as previously described ([OhAinle et al., 2018](#) [↗](#)). For HIV-1 production, 293Ts were transfected with 1 μ g/well HIV_{LAI} or HIV-1_{NL4.3-BaL} proviral DNA or 1 μ g/well HIV_{LAI} Δ env DNA and 500 ng/well pMD2.G for HIV-1_{LAI}, HIV-1_{NL4.3-BaL}, and HIV-1_{LAI}-VSVG, respectively. One day post-transfection, media was replaced. Two days post-transfection, viral supernatants were collected and filtered through a 20 μ m filter and aliquots were frozen at -80°C . HIV-1_{LAI}, HIV-1_{NL4.3-BaL} and HIV-1_{LAI}-VSVG proviruses were previously described ([Bartz and Vodicka, 1997](#) [↗](#); [Gummuluru et al., 2003](#) [↗](#); [Peden et al., 1991](#) [↗](#)).

Cell-free and cell-to-cell coculture HIV-1 infection

Cell-free infections with HIV-1_{LAI-VSVG} were done as previously described (Kulsuptrakul et al., 2023 [DOI](#)). Subsequent cell death was assessed by incubating in media containing propidium iodide dye (10µg/mL) for 5 minutes at room temperature then washed once with PBS before fixing with BD CytoFix/Cytoperm (cat:BDB554714) and staining for intracellular p24^{gag} (Beckman Coulter cat#: 6604665) for flow cytometry. In the HIV-1 cell-to-cell transmission system, SUPT1 expressing CCR5 (SUPT1-CCR5) cells were spinoculated at 1100g for 30min with either HIV-1_{LAI} or HIV-1_{NL4.3-BaL} in the presence of 10µg/mL DEAE-dextran. SUPT1-CCR5 cells were lentiviral transduced to express CCR5 (Dingens et al., 2017 [DOI](#)). After 24 hours, mock or HIV-1 infected SUPT1-CCR5 cells were washed three times in PBS such that DEAE-dextran and cell-free virus were removed before starting coculture with THP-1 cells or MDMs. THP-1 cells and MDMs were seeded at 5×10^5 cells/well and primed with 500ng/mL Pam3CSK4 (Invivogen) for 16-24 hours before coculture. Mock or infected SUPT1 cells were seeded at 5×10^5 cells/well. Cultured supernatants from coculture were harvested 48 hours after starting the coculture for the IL-1R reporter assay, which was previously described (Kulsuptrakul et al., 2023 [DOI](#)).

Transwell coculture HIV-1 infection

SUPT1 cells were spinoculated at 1100g for 30min with HIV-1_{LAI} in the presence of 10µg/mL DEAE-dextran. After 24 hours, mock or HIV-1 infected SUPT1 cells were washed 3 times in PBS and either mixed in a 24-well with THP-1 cells or placed in a transwell insert above target THP-1 cells at a concentration of 5×10^5 infected SUPT1 cells and 2.5×10^5 THP-1 cells per well. THP-1 cells were primed overnight with 500ng/mL Pam3CSK4 before starting coculture. The transwell insert has a 0.4µm membrane at the bottom of the well (ThinCert™ Tissue Culture Inserts, Sterile, Greiner Bio-One cat:665640), allowing virus to pass out of the transwell but not the infected cell. Reverse transcriptase activity in viral supernatants was measured using the RT activity assay as previously described (Roesch et al., 2018 [DOI](#); Vermeire et al., 2012 [DOI](#)). A stock of HIV-1_{LAI} virus was titered multiple times, aliquoted at -80°C and used as the standard curve in all assays.

Monocyte-derived macrophage isolation, differentiation, and editing

Primary monocytes were isolated via negative selection using the EasySep™ Human Monocyte Isolation Kit (Easy Sep, 1×10^9) (Stem Cell Technologies) according to the manufacturer's protocols from PBMCs collected from healthy donors. Upon isolation, monocytes were seeded at 5×10^5 cell/well in 24-well plates and differentiated for 5 days in the presence of media containing 20ng/mL GM-CSF (PeproTech cat: 300-03) and 10ng/mL M-CSF (PeproTech cat: 300-25), changing media every other day. For edited MDMs, isolated monocytes were electroporated in cuvettes (100µL) with $2.5\text{--}5 \times 10^6$ cells/nucleofection in the presence of pre-complexed Cas9-RNPs (600pmol sgRNA: 100pmol Cas9) in Lonza P2 buffer using pulse code DK-100. RNPs were complexed with sgRNA from the Synthego gene KO kit, which includes 3 sgRNAs per gene. Thus, each sgRNA was present at a 2:1 ratio with Cas9 (QB3 MacroLab). A table of sgRNAs used for AAVS1 and CARD8 KO can be found in [Table S2](#) [DOI](#) below. After nucleofection, cells were supplemented with 900µL of prewarmed media and allowed to recover for 30 minutes at 37°C before seeding in 48-well plates at $\sim 6 \times 10^5$ cells/well for differentiation. Media was changed 24 hours post nucleofection then differentiated for 5 more days before characterizing knockout efficiency and conducting coculture experiments.

CARD8 cleavage assay

HEK293T cells were seeded at $1\text{--}1.5 \times 10^5$ cells/well in 24-well plates the day before transfection using TransIT-LT1 reagent at 1.5 µL transfection reagent/well (Mirus Bio LLC). One hundred ng of indicated constructs encoding an N-terminal mCherry-tagged CARD8 were co-transfected into

sgRNA	Sequence
CARD8 sgRNA1	CUCUGCAGUGACAUCAAACA
CARD8 sgRNA2	UGACGAUUGCGUUUGGUUCC
CARD8 sgRNA3	AGCGUUUGGUUCCCCACUGC
AAVS1 sgRNA 1	GUUAAUGUGGCUCUGGUUCU
AAVS1 sgRNA 2	ACCCACAGUGGGGCCACUA
AAVS1 sgRNA 3	CCUCCUAGUCUCCUGAUAU

Table S2

HEK293T cells with empty vector (‘-’), HIV_{LAI} or PI-R provirus. To normalize HIV^{gag} expression between HIV-1_{LAI} and the PI-R clones, which are in a different vector backbone, 400ng of HIV-1_{LAI} and 200ng of all PI-R clones were transfected. All conditions were normalized with empty vector to contain the same amount of DNA. Cytoplasmic lysates were harvested 24 hours post-transfection and immunoblotted as previously described (Kulsuptrakul et al., 2023 [DOI](#)).

HEK reconstitution assay

HEK293T cells, which endogenously express CARD8, were seeded at 2.25×10^5 cell/well in 24-well plates the day before transfection using TransIT-LT1 reagent at 1.5uL transfection reagent/well (Mirus Bio LLC). Functional inflammasomes were reconstituted by transfecting in 5ng human CASP1 and 100ng human IL-1 β . To assess the effects of different viral proteases on inflammasome activation, HIV-1_{LAI} or PI-R clones were co-transfected in with CASP1 and IL-1 β . As with the CARD8 cleavage assay, a higher amount of 250ng HIV-1_{LAI} was added relative to the PI-R clones, which were all added at 125ng, to normalize HIV^{gag} expression between the different vector backbones. All conditions were normalized with empty vector to contain the same amount of DNA. Cultured supernatant was harvested 24 hours post-transfection to assay for IL-1 β secretion via IL-1R reporter assay.

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Joint Public Review:

Following up on their previous work, the authors investigated whether cell-to-cell transmission of HIV-1 activates the CARD8 inflammasome in macrophages, an important question given that inflammasome activation in myeloid cells triggers proinflammatory

cytokine release. The data support the idea that CARD8 is activated by the viral protease and promotes inflammation. However, time-course analyses in primary T cells and macrophages and further information on the specific inflammasome involved would further increase the significance of the study.

Strengths:

The manuscript is well-written and the data is of good quality. The evidence that CARD8 senses the HIV-1 protease in the context of cell-to-cell transmission is important since cell-to-cell transmission is thought to play a key role in viral spread in vivo, and inflammation is a major driver of disease progression. Clean knockout experiments in primary macrophages are a notable strength and the results clearly support the role of CARD8 in protease-dependent sensing of viral spread and the induction of IL1 β release and cell death. The finding that HIV-1 strains are resistant to protease inhibitors differ in CARD8 activation and IL1 β production is interesting and underscores the potential clinical relevance of these results.

Weaknesses:

One weakness is that the authors used T cell lines which might not faithfully reflect the efficiency of HIV-1 production and cell-cell transfer by primary T cells. To assess whether CARD8 is also activated by protease from incoming viral particles earlier time points should be analyzed. Finally, while the authors exclude the role of NLRP3 in IL-1 β and the death of macrophages it would be interesting to know whether the effect is still Gasdermin D dependent.

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Author response:

Thank you for the positive and constructive feedback on our manuscript. We appreciate you highlighting the importance of our work advancing our understanding of the molecular etiology of acquired immunodeficiency syndrome (AIDS). To extend and further substantiate the observation that the CARD8 inflammasome is activated in response to viral protease during HIV-1 cell-to-cell transmission, we are in the process of completing additional experiments that are responsive to reviewer feedback, including:

- Primary CD4⁺ T cell to monocyte-derived macrophage (MDM) transmission: We have now repeated the cell-to-cell experiments with HIV-1 transfer from primary CD4⁺ T cells to primary monocyte-derived macrophages, and our findings are consistent with CARD8-dependent IL-1 β release from HIV-1-infected macrophages in this more physiologic context. We are in the process of repeating these experiments with additional donors and will add these results to the revised manuscript.
- Heterogeneity amongst blood donors: We have now repeated the cell-to-cell transfer and CARD8 knockout in MDMs with additional donors. While we continue to observe heterogeneity amongst donors, the key observation that CARD8 is required for inflammasome responses to HIV-1 infection is consistent. We note that some donors, including the one individual reported in the first submission, have markedly diminished CARD8 activity (to both HIV-1 and VbP).
- Time course experiments: We did conduct a time course experiment when initially establishing these assays. We have now repeated these experiments with additional timepoints and in the presence or absence of the RT inhibitor nevirapine. The results of these experiments will be included in the revised manuscript.

- The role of Gasdermin D: We are mostly interested in the release of IL-1 β from the infected macrophages due to its potential contribution to myeloid-driven inflammation in PLWH. To date, there is no evidence that any other pore-forming protein other than GSDMD can initiate IL-1 β release (and pyroptosis) downstream of CARD8. Nonetheless, we will attempt this experiment with the Gasdermin D inhibitor, disulfiram.

We believe these and other experiments will further support the importance of the CARD8 inflammasome in myeloid-driven inflammation in PLWH and look forward to submitting the revision.

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